

11/24 Polynomials



Lecture 18 Polynomials & polynomial division

Eg)

$$\begin{array}{r} x^2+1 \overline{) x^3 - 2x + 1} \\ \underline{-(x^3 + x)} \\ 0 - 3x + 1 \end{array}$$

when can we do this?

$$x^3 - 2x + 1 = \underbrace{x}_{\text{quotient}} (x^2 + 1) + \underbrace{-3x + 1}_{\text{remainder}}$$

$$\mathbb{Z}[x], \mathbb{Q}[x], \mathbb{R}[x], \mathbb{C}[x]$$

$R[x]$ polynomials w/ coeff in a Ring

$R[x]$ ring of polynomials

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in R[x] \text{ where } a_n, \dots, a_0 \in R$$

equality

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 = b_n x^n + b_{n-1} x^{n-1} + \dots + b_0$$

$$\Leftrightarrow a_i = b_i \quad \forall \quad 0 \leq i \leq n$$

addition

$$\sum_{i=0}^n a_i x^i + \sum_{i=0}^m b_i x^i = \sum_{i=0}^{\max(m,n)} (a_i + b_i) x^i$$

where $a_i = 0$ if $i > n$
or $b_i = 0$ if $i > m$

multiplication

$$\left(\sum_{i=0}^n a_i x^i \right) \cdot \left(\sum_{i=0}^m b_i x^i \right) = \sum_{i=0}^{m+n} \left(\sum_{r+s=i} a_r \cdot b_s \right) x^i$$

Eg $\mathbb{Z}_6[x]$ $(3x+1)(2x+1) = \cancel{6} x^2 + 5x + 1$
 $= 5x + 1$

Def A ring is an integral domain if it is a subring of field.

$\mathbb{Z} \subset \mathbb{Q}$ so $\mathbb{Z}$ is an integral domain

Def A ^{commutative} ring is an integral domain if $a \cdot b = 0 \Rightarrow$

$a=0$ or $b=0$ for all $a, b \in R$.

"no zero divisors"

Prop 10.1 1) R is commutative $\Leftrightarrow R[x]$ is commutative

2) R is an integral domain $\Leftrightarrow R[x]$ integral domain

3) $R[x]$ is never a field.

2) If $R[x]$ is an integral domain then look polynomials of degree zero. Let a & b be polynomial of degree zero by assumption $a \cdot b = 0 \Rightarrow a = 0$ or $b = 0$.

But $a, b \in R$ so R must be an integral domain.

If R is an integral domain. Suppose $\exists f, g \in R[x]$

such that $f \cdot g = 0$. & $f \neq 0$ & $g \neq 0$

suppose $f = a_n x^n + \dots$ & $g = b_m x^m + \dots$

$f \cdot g = a_n b_m x^{n+m} + \dots$

but $f \cdot g = 0 \Rightarrow a_n b_m = 0$ but

$a_n \neq 0$ & $b_m \neq 0$

~~because~~ because R is an integral domain

3) Consider $x \in R[x]$. Suppose $\exists f \in R[x]$ such that $x \cdot f = 1$. $f = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$

then $x \cdot f = a_n x^{n+1} + a_{n-1} x^n + \dots + a_0 x$ which has no constant term. ~~*~~ So x has no inverse and $R[x]$ is not a field.

Cor $F[x]$ where F is a field is an integral domain.

Note $R[x]$ an integral domain has property that $\deg(f \cdot g) = \deg(f) + \deg(g)$

Division Algorithm

Let F be a field & $f, g \in F[x]$ with $g \neq 0_F$

Then there exist unique polynomials q & r such that

$$f = qg + r \quad \& \quad \text{either } r = 0_F \text{ or } \deg(r) < \deg(g).$$

$\nearrow$ quotient $\nwarrow$ remainder

Pf Case I $f = 0$ true if $q = 0$ & $r = 0$

Case II Suppose $\deg(f) < \deg(g)$ $q = 0$ & $r = f$

Case III $\deg(f) \geq \deg(g)$

Proof by induction on the degree of f .

Base case $\deg(f) = 0$ then $f = a \in F$ & $g = b \in F$

$$a = (a \cdot b^{-1})b + 0 \quad f = (a \cdot b^{-1})g + 0$$

so $q = a \cdot b^{-1} \quad \& \quad r = 0$

Assume the division algorithm holds for all f with degree strictly less than n .

$$f = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 \quad a_n \neq 0$$

$$g = b_m x^m + b_{m-1} x^{m-1} + \dots + b_0 \quad b_m \neq 0$$

$$\& \quad m \leq n$$

Look at $a_n b_m^{-1} x^{n-m} g = a_n x^n + \dots$

$f - a_n b_m^{-1} x^{n-m} g$ is a polynomial with degree strictly less than n

By induction hypothesis, $\exists q_1 \in R$ such that

$$f - a_n b_m^{-1} x^{n-m} g = q_1 g + r \quad \text{w/ } \deg r < \deg g \text{ or } r = 0_R$$

$$f - a_n b_m^{-1} x^{n-m} g = q_1 g + r \quad \text{w/ } \deg r < \deg g \text{ or } r=0$$

$$f = \underbrace{\left(a_n b_m^{-1} x^{n-m} + q_1 \right)}_{\text{quotient}} g + \underbrace{r}_{\text{remainder}} \quad \text{with } \deg(r) < \deg(g) \text{ or } r=0$$

$$\text{So } q = a_n b_m^{-1} x^{n-m} + q_1 \quad \& \quad r = r$$

Uniqueness of $q \& r$

Suppose $\exists \quad q' \& r'$ such that where $\deg r' < \deg g$ or $r'=0$

$$f = q' \cdot g + r' \quad \text{where } q' \neq q$$

$$qg + r = f = q'g + r' \Rightarrow \underbrace{(q - q')g}_{\text{degree must be at least } \deg g} = \underbrace{(r' - r)}_{\text{degree strictly less than } g}$$

contradiction so $q' = q$ $\leftarrow$

Now

$$f = zg + r$$

$$f = zg + r'$$

$$\Rightarrow \cancel{zg} + r = \cancel{zg} + r'$$

$$\Rightarrow r = r' \quad \square$$